

External Validation of Finding-to-Report Faithfulness Scoring Across Sites and Equipment

A Hidden-Holdout Protocol

Natan Paraiso Ribeiro, MD Petrus Paraiso Ribeiro, MD Stephanie Alba Herrera, MD
Francisco Akira, MD Raquel Moreno, MD

Laudos.AI Research • research@laudos.ai

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Abstract

Public synthetic cases are useful for harness inspection but are weak evidence for clinical generalization: they are contaminable and not drawn from real reporting distributions. This protocol pre-registers an external-validation study for the LAIBench scorer and for the systems it ranks, using a *hidden holdout* that is never published and never used for tuning. We specify the holdout construction, the cross-site/cross-equipment stratification, the frozen-submission and contamination-audit procedures, and the generalization analysis. **No results are reported here**; this document fixes the design before the holdout is built or scored.

1 Motivation

A score on a contaminable public set, or on a controlled set that has been used during development, cannot support a strong external claim. LAIBench therefore separates four tiers: public-smoke (contaminable, demo only), controlled-eval (gated, first-party, aggregate-only), calibration-set (scorer tuning only), and *hidden-holdout* (never seen, never tuned). Only the hidden-holdout can sustain a strong generalization claim. This protocol governs that tier.

2 Design

Holdout construction. A held-out set assembled to span multiple institutions, scanners/equipment vendors, and modalities, disjoint from any case ever used for development or calibration. The holdout is sealed (suite hash recorded) and access-controlled; no case text, answer key, or example from it appears in any public artifact (paper, site, README, prompts, demos).

Frozen submissions. The leaderboard accepts only runs produced by the official runner or validated frozen predictions: each submission JSONL is hashed and timestamped and pinned to a scorer version, with no unlimited resubmission against the same holdout.

Contamination audit. For every run we record the declared training-data cutoff (where known), the public prompt used, the suite hash, a canary-token scan, whether the model/provider may have seen the public repository, and whether outputs echo case IDs or benchmark boilerplate; each run publishes a summary *contamination scan: pass / fail / unknown*.

3 Analysis plan

Report scorer and system performance overall and stratified by site, equipment vendor, and modality; quantify the generalization gap between controlled-eval and hidden-holdout. Pair the hidden-holdout reading with the radiologist-adjudication protocol so that scorer-human agreement is estimated on never-tuned data. Pre-specify the strata, the minimum per-stratum case counts, and the degradation threshold beyond which a generalization claim is withheld.

4 Governance and limitations

The golden rule is enforced operationally: any suite used for debugging or scorer tuning is retired as a holdout. This study measures generalization and contamination control; it does not by itself establish clinical safety or regulatory fitness, and it is not image interpretation. Until it reports, LAIBench claims remain first-party and aggregate, with external validation explicitly *pending*.